An entry–exit law for the resurrection of dead neurons under smoothed ReLU activation

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July 25, 2026

Abstract

A neuron dies when its pre-activation margin is negative on every datum; under a hard ReLU it then has zero gradient and never returns. We ask what replaces this under smoothed activations. For a shallow softplus network with scalar input and unit output weights under gradient flow we prove that, in the sharp-activation limit, the dynamics is a hybrid system with memory: the dead unit freezes, and reactivates exactly when the signed time integral of the residual excluding it returns to zero, the classical entry–exit function, with a universal first-order release coefficient; all constants are verified on one two-neuron instance. The two limit orders do not commute: the pointwise generalized-gradient flow freezes dead units forever, the smoothed flow revives them. A pre-registered experimental ladder follows: a reduced stochastic-schedule model is falsified, two instrument designs are recorded as negatives, and a confirmatory MLP run (400 runs, noiseless gradient descent) reproduces resurrection in half of the deaths observable within the horizon ($p = 0.50$, 15/30; $p = 0.457$ over all 35 deaths), the area opening before 13 of 16 returns, hard ReLU giving a structural zero, and return probability monotone in the stiffness. Transfer to a transformer feed-forward block is a pre-registered protocol, not established here.

Keywords: entry–exit function; geometric singular perturbation; dead neurons; smoothed ReLU; gradient flow; loss of plasticity.

MSC 2020: Primary 34E15; Secondary 34A36, 68T07.

1 Introduction

Write $\Phi : \mathcal{W} \to \mathcal{F}$ for the parametrization of a network and $J = D\Phi$ for its Jacobian. A neuron whose pre-activation margin is negative on every datum contributes an identically zero column to J: it sits on a stratum of the singular locus where J drops rank. Under a hard ReLU the gradient of the loss in the directions of that unit is exactly zero, so the stratum is absorbing. A dead unit stays dead. This is the mechanism behind the empirically documented dormant-neuron and loss-of-plasticity phenomena [1, 2, 3]. The question of this

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paper is what replaces the absorbing stratum when the activation is smoothed, as it is in every softplus or GELU [4] network actually trained.

Our answer is an exact gating law and its numerical arc. The contributions are:

- **A theorem (Section 3 and Theorem 2).** In the sharp limit $\beta \to \infty$ of a shallow softplus network under gradient flow, the death and resurrection of a neuron is the entry–exit problem of a slow–fast system [5]: the dead unit freezes and reactivates when $\int R\,dt = 0$, where R is the residual excluding that unit. The uniform rate is $O(\log\beta/\beta)$ and the release time carries a universal first-order coefficient $-1/(cR(t_{\text{exit}}))$. The proof is organized as six lemmas (Section A); each carries, in addition, an independent executable symbolic or numerical verification, which is not standard practice and which we therefore state explicitly. For the erf smoothing we verify numerically that the release criterion is unchanged; the proof is written for softplus, and GELU is outside the class because its derivative is non-monotone.

- **Non-commutation (Section 4).** The two limit orders, smooth-then-sharp against the pointwise generalized-gradient flow, do not commute. The field-first order is the frozen branch of Eberle, Jentzen, Riekert and Weiss [6]; our flow-first limit resurrects. This is the exact mathematical content of "dead ReLU units stay dead, smooth units come back."

- **A pre-registered experimental ladder (Section 6).** Organized by prediction, with the null controls and the negatives shown: a reduced deterministic-schedule model falsified under SGLD (Section 6.1); two instrument designs recorded as negatives because a calibrated null fired (Section 6.2); the mechanism first observed in an MLP pressure pilot (Section 6.3); and a confirmatory MLP run of 400 runs in which the area law reproduces with confidence intervals, hard ReLU acts as the zero-gradient control, and the return probability is monotone in β (Section 6.4 and Figure 4). The scoping of the transformer (Section 6.6) exposed two new mechanisms, a GELU sign-flip barrier and an upstream channel, and the corresponding confirmatory is stated here as a pre-registered protocol.

The theorem covers one two-neuron instance; the confirmatory covers one small architecture at zero noise; the stochastic regime is open and, where we probed it, it breaks the deterministic schedule (Section 6.1).

The paper is organized as follows. Section 2 fixes the model and the four hypotheses. Section 3 states the entry–exit theorem, the six lemmas that compose it and the numerical verification of its constants; the complete proofs of the lemmas are deferred to Section A. Section 4 makes the non-commutation of the two limit orders precise and connects it to the plasticity literature, which Section 5 surveys. Section 6 is the experimental ladder, from the falsified reduced model to the confirmatory and to the transformer protocol; Sections 7 to 9 give the limitations, the conclusion, and the reproducibility record.

2 Setup and notation

Data $(x_i, y_i)_{i=1}^n \in \mathbb{R} \times \mathbb{R}$ in general position; shallow network with unit output weights

$$f_\theta(x) = \sum_k g_\beta(w_k x + b_k), \qquad g_\beta(z) = \beta^{-1} \log(1 + e^{\beta z}), \tag{1}$$

quadratic loss $L = \frac{1}{2n} \sum_i (f_\theta(x_i) - y_i)^2$ and gradient flow $\dot\theta = -\nabla L$ on the training horizon $[0, H]$. General monotone sigmoids are treated in Section 3. The parameter β is the *activation stiffness*: as $\beta \to \infty$, $g_\beta \to \mathrm{ReLU}$ and $g'_\beta \to \mathbf{1}_{\{z>0\}}$. We write H for the training horizon throughout, a time in the flow and a number of optimizer steps in the experiments, and $\mathcal{N}$ for the standard normal cumulative distribution function.

Fix one neuron A and one datum j. Its *margin* is $z = \theta_A \cdot (x_j, 1)$; write $c = \|(x_j, 1)\|^2/n$ and let

$$R(t) = y_j - f_{-A}\big(x_j; \theta_B(t)\big) \tag{2}$$

be the *residual excluding* A, where θ_B collects the other parameters. The *hybrid reference* θ^{hyb} runs the full ReLU flow until the first time t_{hit} with $z = 0$, then freezes θ_A and runs θ_B alone; set R^{hyb} accordingly and

$$\mathcal{A}^{\mathrm{hyb}}(t) = \int_{t_{\mathrm{hit}}}^{t} R^{\mathrm{hyb}}(s)\, ds \tag{3}$$

for its signed area. The death of A is exactly the rank drop of J on the stratum $\{z \le 0\}$; the question is the fate of the trajectory on that stratum.

Hypotheses. On the episode we assume

$R^{\mathrm{hyb}}(t_{\mathrm{hit}}) < 0$ (transversal death); (H1)

$\mathcal{A}^{\mathrm{hyb}}$ has a first zero at $t_{\mathrm{exit}} < H$ with $R^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$ and $\mathcal{A}^{\mathrm{hyb}} < 0$ on $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$; (H2)

the other margins of A stay $\le -\delta_1 < 0$; (H3)

no other switching event occurs in $[0, H]$. (H4)

3 The entry–exit theorem

Definition 1 (hybrid limit)**.** θ^∞ equals: the ReLU flow on $[0, t_{\mathrm{hit}}]$; on $[t_{\mathrm{hit}}, t_{\mathrm{exit}}]$, $\theta_A \equiv \theta_A(t_{\mathrm{hit}})$ (frozen) with θ_B the B-only flow; on $[t_{\mathrm{exit}}, H]$, the ReLU flow restarted at $(\theta_A(t_{\mathrm{hit}}), \theta_B^{\mathrm{hyb}}(t_{\mathrm{exit}}))$.

Theorem 2 (entry–exit law for the smoothed dead unit)**.** *Assume* (H1)–(H4)*. Then for every* $\delta > 0$

$$\sup_{t \in [0,H] \setminus (B_\delta(t_{\mathrm{hit}}) \cup B_\delta(t_{\mathrm{exit}}))} \big\|\theta^\beta(t) - \theta^\infty(t)\big\| = O_\delta\!\left(\frac{\log \beta}{\beta}\right), \tag{4}$$

and the first return $t^\beta_{\mathrm{release}}$ *of the margin* z *to* 0 *after death satisfies*

$$t^\beta_{\mathrm{release}} = t_{\mathrm{exit}} - \frac{1}{c\, R^{\mathrm{hyb}}(t_{\mathrm{exit}})} \cdot \frac{1}{\beta} + o(\beta^{-1}). \tag{5}$$

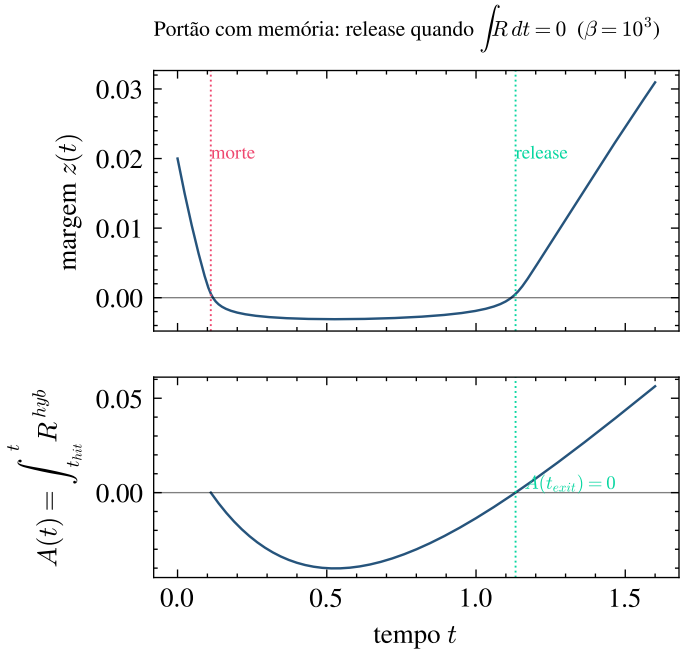


Figure 1: The memory gate at $\beta = 10^3$ (coupled system, reference instance). Top: the margin $z(t)$ dies at t_{hit}, hugs the sticking layer of depth $\sim \beta^{-1}\log\beta$, and releases. Bottom: the hybrid area $\mathcal{A}(t) = \int_{t_{\mathrm{hit}}}^{t} R^{\mathrm{hyb}}$; release coincides with $\mathcal{A} = 0$ up to $O(\beta^{-1})$ (Theorem 2). Deterministic Euler integration ($dt = 10^{-5}$); the uncertainty is discretization only. Script `t2-prova/figuras_draft.py` (committed).

Remark 3 (what carries each constant). The $\log\beta$ in (4) is sharp and is carried solely by the normal coordinate of θ_A inside the episode, whose sticking depth is $z \sim -\beta^{-1}\log\beta$ (Lemma 8); θ_B obeys $O(\log\beta/\beta^2)$ (Lemma 6) and the tangential components of θ_A are exponentially frozen (Lemma 10). The first-order coefficient in (5) is universal: in the product of the entry delay by the area it cuts, the arrival speed cancels, leaving the reciprocal exit speed, so each corner contributes the reciprocal of its own transversal speed (Lemma 9). The remainder is $o(\beta^{-1})$; the numerics suggest $O(\log\beta/\beta^2)$, which would follow from quantifying the $o(1)$ of Lemma 7 with $\delta_z = \log\beta/\beta$, and we leave it as a stated refinement. The limit is that of the regularized flow; discrete ReLU gradient descent freezes the dead neuron forever (Section 4).

Remark 4 (which hypotheses are essential). (H1)–(H2) are essential: they are the entry–exit geometry itself, and the law fails without the sign change of R^{hyb}. (H3)–(H4) are there to simplify the proof: (H3) makes the other data contribute $O(e^{-\beta\delta_1})$ and (H4) isolates a single episode. We expect both to be removable at the cost of bookkeeping; we have not done it.

3.1 The six lemmas

The episode reduces to a scalar entry–exit problem through the six lemmas below, proved in Section A. Each also carries a committed executable symbolic or numerical verification (`passo0--6`, `prova_L1--L6`), whose outcome is reported at the end of each proof.

Lemma 5 (normal form and additive quadrature)**.** *Let $w = e^{\beta z}$ and $u = 1/w$, and set $\Psi(u) = \log u + u$. In fast time $\tau = \beta t$ the margin obeys*

$$\frac{dw}{d\tau} = c\,R(t)\,\frac{w^2}{1+w}\,(1+E_\beta), \qquad |E_\beta| \le \frac{\log(1+w)}{\beta|R|}, \tag{6}$$

the quadratic normal form of De Maesschalck and Schecter [5, eqs. (1.9), (1.10)] with $\varepsilon = \beta^{-1}$, and one has the exact additive identity

$$\frac{d\Psi(u)}{dt} = -c\,\beta\,R(t) + c\log(1+w) + \mathcal{E}_{(\mathrm{H3})}(t), \qquad \big|\mathcal{E}_{(\mathrm{H3})}\big| \le C_3\,\beta\,e^{-\beta\delta_1}. \tag{7}$$

Under (H1)–(H2) the additive correction satisfies $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{release}}} c\log(1+w)\,ds = O(\log\beta/\beta)$.

Lemma 6 (B-subsystem and feedback)**.** *Under (H4) the B-field is affine with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, where G_B is the Gram matrix of the active data, and R^{hyb} is affine in θ_B. Integral Grönwall consuming the bound of Lemma 5 gives, conditionally on the layer being maintained (supplied by Lemma 9),*

$$\sup\big\|\theta_B^{\beta} - \theta_B^{\mathrm{hyb}}\big\| = O\!\left(\frac{\log\beta}{\beta^2}\right), \tag{8}$$

one full order better than the naive $O(\beta^{-1})$: in the bulk $u = O(\beta|\mathcal{A}|)$, so the coupling is $O(\beta^{-2})$ there and $O(\beta^{-1})$ only in the $O(\beta^{-1})$-wide corner windows.

Lemma 7 (entry corner)**.** *Let $v = c\,|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|$ be the arrival speed, transversal by (H1). The exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$ collapses the crossing-time difference to an elementary integral and gives*

$$t_{\mathrm{cross}}^{\beta} = t_{\mathrm{hit}} + \frac{1}{\beta v}\big(1 + O(\delta_z) + o(1)\big), \qquad \big\|\theta^{\beta}(t_{\mathrm{cross}}) - \theta^{\infty}(t_{\mathrm{hit}})\big\| = O(\beta^{-1}), \tag{9}$$

where $t_{\mathrm{cross}}^{\beta}$ is the first time with $w = w_0$. The exponentially weighted approach integral is $\frac{3/2}{\beta v}(1+o(1))$, the factor $3/2$ coming from $\int_0^\infty e^{-s}(1+e^{-s})\,ds$.

Lemma 8 (sticking and release for prescribed R)**.** *Let $R \in C^1$ be prescribed and satisfy (H1)–(H2), and write $\mathcal{A}(t) = \int_{t_{\mathrm{hit}}}^{t} R$. Then*

(i) $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + \mathrm{corr}(t)$ with $0 \le \mathrm{corr}(t) = \int c\log(1+w)\,ds = O(\log\beta/\beta)$; on compact subintervals of $(t_{\mathrm{hit}}, t_{\mathrm{exit}})$, $u = c\beta|\mathcal{A}(t)|\big(1 + O(\log\beta/(\beta a_0))\big)$, so the sticking depth is $z \sim -\beta^{-1}\log\beta$;

(ii) the release obeys the closed formula

$$t_{\text{release}} - t_{\text{exit}} = \frac{\operatorname{corr}(\beta)}{c\,\beta\,R(t_{\text{exit}})}\,(1+o(1)) = O\!\left(\frac{\log\beta}{\beta^2}\right), \tag{10}$$

so u returns to $O(1)$ at $t_{\text{exit}} + O(\beta^{-1})$, with no $\log\beta$ factor;

(iii) a perturbation $\tilde{R} = R + \delta$ shifts t_{exit} by at most $\delta\,(t_{\text{exit}} - t_{\text{hit}})/R(t_{\text{exit}}) + O(\delta^2)$.

No division by R occurs anywhere: the layer is held by the area, not by the sign of R, and the interior zero of R forced by (H1)–(H2) is harmless.

Lemma 9 (coupled bootstrap)**.** *Under (H1)–(H4) the coupled system satisfies the two-term law*

$$t^{\beta}_{\text{release}} = t_{\text{exit}} - \frac{1}{c\,R^{\text{hyb}}(t_{\text{exit}})}\cdot\frac{1}{\beta} + \frac{\operatorname{corr}(\beta)}{c\,\beta\,R^{\text{hyb}}(t_{\text{exit}})} + o(\beta^{-1}), \qquad \operatorname{corr}(\beta) = O\!\left(\frac{\log\beta}{\beta}\right), \tag{11}$$

obtained by continuation with strict margins, with the constant $M_2 = e^{L_B H} C_f C_4/c$ independent of the continuation hypothesis for $\beta \ge \beta_0$. The first-order coefficient is universal: in the product $K\,|R^{\text{hyb}}(t_{\text{hit}})|/R^{\text{hyb}}(t_{\text{exit}})$ of the entry delay by the area it cuts, the factor $|R^{\text{hyb}}(t_{\text{hit}})|$ cancels against $K = 1/(c|R^{\text{hyb}}(t_{\text{hit}})|)$ of Lemma 7.

Lemma 10 (exit corner, branch selection and scalar reduction)**.** *Under (H1)–(H4), after release: (i) the β-trajectory crosses the exit corner in time $\frac{1}{\beta v'}(1+o(1))$ with $v' = c\,R^{\text{hyb}}(t_{\text{exit}}) > 0$, the mirror of Lemma 7; (ii) the ReLU flow restarted at $(\theta_A(t_{\text{hit}}), \theta_B^{\text{hyb}}(t_{\text{exit}}))$ has forward uniqueness on the activating branch, and that branch is selected by the limit itself, the frozen Filippov branch [7] being incompatible with release at $t_{\text{exit}} + O(\beta^{-1})$; (iii) classical Grönwall glues on $[t_{\text{release}}, H]$ at rate $O(\log\beta/\beta)$. Under (H3) the other data contribute $O(e^{-\beta\delta_1})$, the face-j field is rank one along $(x_j, 1)$, and the episode reduces to the scalar problem.*

3.2 Proof of the theorem

Proof of Theorem 2. We compose the six lemmas along the four stages of the episode.

By Lemma 10 the field on θ_A splits under (H3) into a rank-one part along $(x_j, 1)$ and terms of size $O(e^{-\beta\delta_1})$, so the tangential components of θ_A are frozen at an exponential rate and the normal component is exactly the scalar problem of Lemmas 5 to 9.

On $[0, t_{\text{hit}} - \delta]$ both flows are the ReLU flow up to the softplus defect, which is bounded by $Ce^{-\beta z^0(t)}$; step 1 of the proof of Lemma 7 integrates that defect against the trajectory and gives a gap $O(\beta^{-1})$. Lemma 7 then delivers the entry corner: the crossing time is $t_{\text{hit}} + 1/(\beta v)(1+o(1))$ and the state at crossing is $O(\beta^{-1})$ from $\theta^{\infty}(t_{\text{hit}})$.

On $[t_{\text{hit}} + \delta, t_{\text{exit}} - \delta]$, Lemma 9 closes the feedback loop: its continuation argument keeps the margin strictly inside the layer and θ_B within $O(\log\beta/\beta^2)$ of the hybrid, by consuming Lemma 6 (which needs the layer) and Lemma 8(iii) (which tolerates the resulting perturbation of the forcing). Inside the layer Lemma 8(i) gives the depth $z \sim -\beta^{-1}\log\beta$, and this is the largest of the three contributions to (4); θ_B contributes $O(\log\beta/\beta^2)$ and the tangential part of θ_A contributes $O(e^{-\beta\delta_1})$. Hence the supremum over this stage is

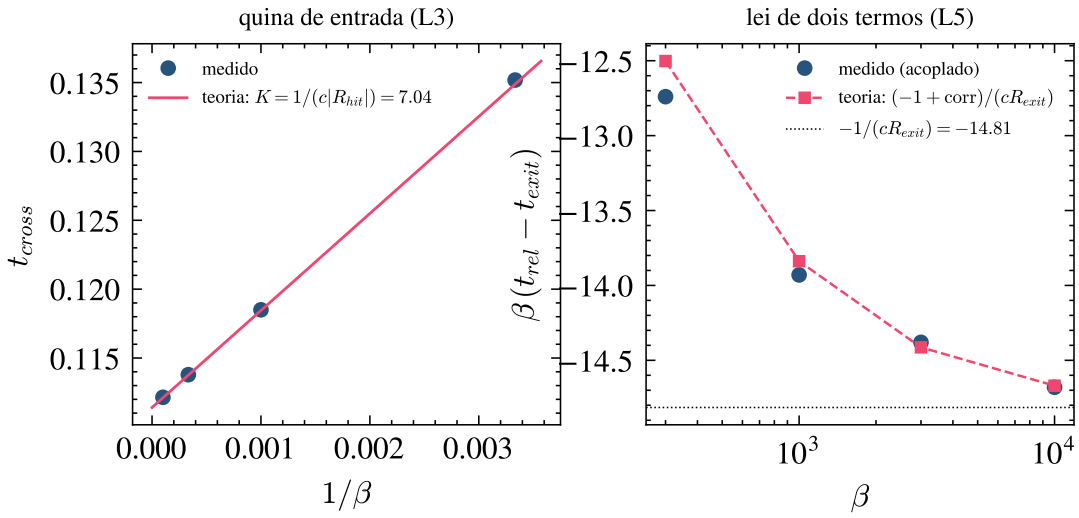


Figure 2: Predicted constants against measurements (deterministic runs; the uncertainty is discretization). Left: entry-corner crossing times against $1/\beta$; the line is the proved slope $K = 1/(c|R^{\mathrm{hyb}}(t_{\mathrm{hit}})|) = 7.04$ (Lemma 7), not a fit. Right: the two-term release law (Lemma 9, squares) against the coupled measurements (circles) over $\beta \in [3 \cdot 10^2, 10^4]$; the dotted line is the universal first-order coefficient $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = -14.81$. Script `t2-prova/figuras_draft.py`.

$O_\delta(\log\beta/\beta)$, carried by the normal coordinate alone, which proves the sharpness asserted in Remark 3.

The terminal window and the release law are Lemma 9 step 5: the correction delay of Lemma 8(ii) is $O(\log\beta/\beta^2)$, the perturbation of the forcing shifts t_{exit} by $O(\log\beta/\beta^2)$ by Lemma 8(iii), and the area cut by the entry corner advances the exit by $-1/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})\beta)$. Summing the three gives (11), hence (5).

Finally Lemma 10 crosses the exit corner in time $O(\beta^{-1})$, selects the activating branch, and glues the two flows on $[t_{\mathrm{release}}, H]$ at rate $O(\log\beta/\beta)$. Taking the maximum of the four stage bounds gives (4). □

3.3 Numerical verification on one instance

The reference instance is $X = (0.5, 1.5)$, $Y = (0.5, 1.0)$, $\theta_A(0) = (-0.5, 0.27)$, $\theta_B(0) = (1.144, 0.261)$, giving $t_{\mathrm{hit}} = 0.1114$, $t_{\mathrm{exit}} = 1.1325$ and $\delta_1 = 0.508$. Every empirically measured constant is predicted by the proofs: the entry coefficient is $K_{\mathrm{teo}} = 7.04$ against 7.13 fitted (1.2%); the coupled two-term law $(-1 + \mathrm{corr})/(cR_{\mathrm{exit}})$ reproduces the fine-grid coupled sweep $-12.74/-13.93/-14.38/-14.68$ to 1.9%/0.7%/0.2%/0.1% for $\beta = 3 \cdot 10^2 \ldots 10^4$, with universal coefficient $1/(cR_{\mathrm{exit}}) = 14.815$ (against -14.68 the leading coefficient alone is

at 0.9%); the tangential drift is at machine zero. All numerics are this single two-neuron instance, and breadth is future work.

3.4 Independence of the smoothing

The release criterion depends only on $\int R\,dt = 0$ and not on the sigmoid. For the erf class, $g'_\beta = \mathcal{N}(\beta z)$, the coupled system releases at the same t_{exit} with an $O(\beta^{-1})$ gap; this is verified numerically and not proved. GELU itself, $z\,\mathcal{N}(z)$, has a non-monotone derivative and is not a member of the family for any β; Section 6.6 shows that this is a distinct mechanism rather than a technicality.

4 Non-commutation and the plasticity bridge

Theorem 2 is the flow-first limit: one takes $\beta \to \infty$ of trajectories. The field-first order instead flows the pointwise-limit generalized gradient, and it freezes degenerate neurons forever. That branch is the rigorous one of Eberle, Jentzen, Riekert and Weiss [6, Theorem 3.3], whose selected solution is unique and has a degenerate set that is non-decreasing in time; their degenerate stratum is the fully-zero one rather than our functionally dead neuron, but the convention $\mathrm{ReLU}'(0) = 0$ and the no-revival mechanism are the same.

Proposition 11 (the two limit orders do not commute)**.** *Let L_∞ be the loss of the hard-ReLU network and let $\nabla^0 L_\infty$ be its pointwise-limit gradient, with the convention $\mathrm{ReLU}'(0) = 0$; let θ^{FF} solve $\dot\theta = -\nabla^0 L_\infty(\theta)$ from $\theta^{\mathrm{hyb}}(t_{\mathrm{hit}})$, which exists and is unique by [6, Theorem 3.3]. Assume (H1)–(H4). Then $\theta_A^{\mathrm{FF}} \equiv \theta_A(t_{\mathrm{hit}})$ on $[t_{\mathrm{hit}}, H]$, whereas θ^∞ of Definition 1 releases A at $t_{\mathrm{exit}} < H$. The two limits therefore disagree on $(t_{\mathrm{exit}}, H]$.*

Proof. The A-block of $\nabla^0 L_\infty$ at θ is $-\frac{1}{n}\sum_i \mathrm{res}_i\,\mathbf{1}_{\{z_{A,i}>0\}}(x_i, 1)$, which vanishes identically whenever every margin of A satisfies $z_{A,i} \le 0$. At t_{hit} the margin on datum j is zero and, by (H3), the remaining margins are $\le -\delta_1 < 0$, so the A-block vanishes and $\dot\theta_A^{\mathrm{FF}} = 0$. The margins of A are functions of θ_A alone, so the condition $z_{A,i} \le 0$ for all i is preserved as long as θ_A is constant, and the set of such θ is therefore forward invariant: θ_A^{FF} stays at $\theta_A(t_{\mathrm{hit}})$ on all of $[t_{\mathrm{hit}}, H]$. By Definition 1 and (H2), θ^∞ restarts the active flow at $t_{\mathrm{exit}} < H$ with $\dot z(t_{\mathrm{exit}}^+) = cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, so $z > 0$ immediately after t_{exit}. □

In machine-learning terms this non-commutation is the statement behind an empirical trend. Discrete ReLU units that die stay dead, which is the dormant-neuron and loss-of-plasticity phenomenon documented across deep and reinforcement learning [1, 2, 3] and mitigated in practice by recycling dead units or by migrating to smooth activations; the smoothed flow resurrects them by the area law. The channel that reopens a smoothed unit is $e^{\beta z}$, which closes as $\beta \to \infty$. Section 6.4 measures the resulting staircase directly.

5 Related work

Entry–exit and geometric singular perturbation. The release criterion $\int R\,dt = 0$ is the classical entry–exit function of slow–fast systems [5, Theorem 1.1], applied for instance to a neurotransmitter release model in [8], and Lemma 5 places our episode in the quadratic normal form of [5, eqs. (1.9), (1.10)] with $\varepsilon = \beta^{-1}$. Our switching face is degenerate in a way that literature does not cover: it carries an identically zero field on the dead side. Generic piecewise-smooth exit theory [9, 10, 11] assumes a nonzero one-sided field, and the canonical Sotomayor–Teixeira regularization [12, 13] requires both one-sided normal components to be nonzero. The degeneracy is of the exponentially flat type treated by Kristiansen [14] with the blow-up method of Krupa and Szmolyan [15], but there the forcing is autonomous, whereas here the residual is generated by the very trajectory it gates. We are not aware of an analogue of that coupling in the entry–exit literature.

Gradient flows of ReLU networks. Existence, uniqueness and convergence rates for gradient flows of ReLU networks are established in [6], whose selected solution is the field-first branch of Proposition 11: once a neuron is degenerate it stays degenerate. Our result is the complementary, flow-first, selection, and the two disagree.

Dormant neurons and loss of plasticity. Dead or dormant units are a measured pathology of trained networks: the dormant neuron phenomenon in deep reinforcement learning [1], the loss of plasticity under distribution shift [2], and its persistence in deep continual learning [3]. The practical mitigations are recycling dead units and smoothing the activation. Theorem 2 and Proposition 11 say what the second mitigation buys: a channel of width $e^{\beta z}$ and a return time set by an area.

6 Experiments

The claims are tested by prediction, each with its null control, and the failed designs are reported alongside the successful ones. Every confirmatory analysis was frozen by a pre-registration commit before the run, and hypotheses that fell were archived rather than rewritten. The commits are: pressure pilot `b48b527`; rate pilot `995a87c`; 4a.3 pre-registration `a8f8f8c`; 4a.3 confirmatory `5a961cb`; transformer pre-registration `1b45498`.

Statistics. Proportions are reported as a point estimate with a two-sided 95% Clopper–Pearson interval [16]; the variability captured is the Bernoulli count of returns among deaths within a single arm, with the denominator given in each cell. Event rates λ, dynamical deaths per run, are reported with a 95% percentile bootstrap interval over $B = 2 \cdot 10^4$ resamples clustered by seed, so that the variability captured is seed-to-seed and not episode-to-episode. Nothing is pooled across arms. We write T for the SGLD temperature, that is, the scale of the injected noise; $T = 0$ is deterministic gradient descent, and H remains the training horizon in optimizer steps.

6.1 The deterministic schedule is falsified under SGLD

Before any confirmatory claim we tested whether the deterministic entry–exit schedule survives noise. A reduced "race" model, pre-registered at commit `190d94a` after two adversarial-review rounds, predicts the capture (resurrection) probability of a dead unit from the mean residual schedule; the falsifying run of 2700 runs under SGLD [17] rejected it. The predicted-zero window was violated decisively: 139 late captures occurred where the model gives exactly zero, and the bin $[0, 0.1)$ saw 150 captures against 1.4 predicted (Poisson $z = +131$ against the model's own expected count, under the model's own independence assumption). The model is well calibrated where it says capture is likely, in the $(0.5, 0.9)$ bin, with $z = -0.26$; the entire error is on the "impossible" side. The theorem-part of the model held: in all nine cells the observed censoring of 2–7% matched the censoring rate that the reduced model itself predicts, from a Brownian first-passage approximation to the margin inside the cone of the deterministic schedule. The failure is therefore of the hypothesis, a deterministic schedule, and not of the instrument.

A follow-up note records what happened next. The natural fix, a stochastic Ornstein–Uhlenbeck schedule with no free parameter, predicts p(capture) nearly identical to the deterministic one, with differences ≤ 0.013, and a late-capture rate of 0.1–1.5% against the $\sim 13.8\%$ observed, which is insufficient by an order of magnitude. That model was therefore refuted at the predictor stage and never run; the leading candidate became the back-reaction of the reactivating unit on the shared residual. The stochastic gate remains open and is out of scope for the theorem, whose regime is deterministic gradient flow.

6.2 Two instrument designs failed their own null control

The first attempt to see the law in a real MLP failed on the instrument, and the failure was caught by a pre-registered null (the first MLP stage, labelled `etapa-4a` in the repository; commit `6411ac8`, 600 runs, MLP with $m \in \{16, 64, 256\}$, SGLD with $\eta = 3 \cdot 10^{-4}$). The pure-diffusion arm passed the calibrated test ($m = 16$, $p = 0.002$); by the pre-registered rule the ruler is then biased and no scientific verdict on the substantive predictions may be claimed, so the run is invalid. The diagnosis is in Figure 3: the ruler was measuring boundary chatter rather than the physical death of the theorem. Of all episodes 59% lasted ≤ 5 steps, and hard ReLU "resurrected" at the same rate and with the same duration distribution as softplus at $T > 0$, which is impossible by gradient, since a dead ReLU unit has exactly zero gradient; these were noise threshold-crossings. The real physics lives at $T \to 0$: the only genuinely long episodes, with median ~ 1110 steps, sit in the $T = 0$ arm and are 99.6% censored at the 10^5-step horizon. The calibrated null held its false-positive rate near α in i.i.d. worlds, but the real world has per-neuron heterogeneity.

A redesign, `etapa-4a.2`, was itself rejected at the design stage by the same process (commit `a413b19`): at $T = 10^{-5}$ the ReLU-to-softplus ratio stayed ≈ 0.95 after debounce, because gradient starvation at $\beta = 300$ masked the effect, and the pooled null broke α by 30–92$\times$ in heterogeneous and autocorrelated worlds. The inheritance was explicit: a hysteresis and dwell ruler, a per-neuron stratified null, and a move to the deterministic

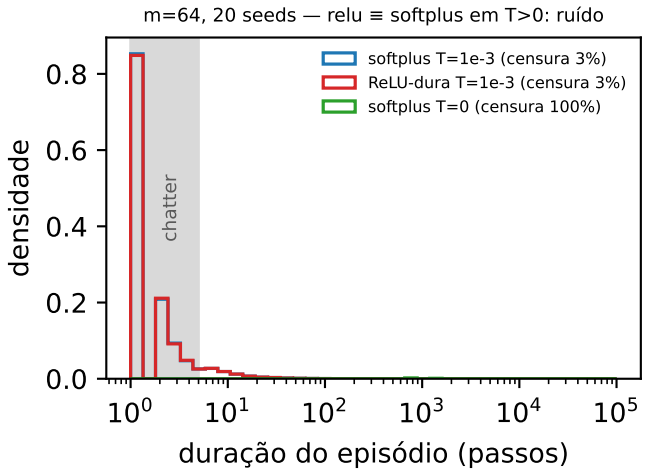


Figure 3: Why `etapa-4a` is an instrument negative (post-hoc diagnostic, $m = 64$, 20 seeds). Episode-duration densities: at $T > 0$ hard ReLU coincides with softplus (median ~ 4 steps, shaded chatter band), a signature of noise rather than of the zero-gradient theorem; the real physics is the $T = 0$ arm ($\sim 10^3$ steps, 99.6% censored). Script `etapa4-mlp-populacao/figuras_diagnostico.py`.

regime $T = 0$ with a much longer horizon. These negatives falsify neither the area law nor the toy calibration. They falsify two rulers, and they are what forced the working design.

6.3 A softplus unit dies and returns in a real MLP, and the area opens first

Moving to $T = 0$ and to a sub-parametrized regime made the mechanism visible. The regime is the *pressure* pilot: a plateau at ~ 0.15 on the rich target $y = \sin(2\pi x) + 0.3\sin(5\pi x)$ with $n = 16$, $\eta = 10^{-3}$ and $H = 2 \cdot 10^5$; 64 runs, commit `b48b527`, exploratory. Hard ReLU gave a structural zero: 0 resurrections out of 25 deaths across 16 seeds, so the frozen branch of Proposition 11 is exact in a real network at $T = 0$. Softplus resurrected by mechanism: at $\beta = 10$ a unit died at step 781 and returned at 139,179; at $\beta = 300$ one died at 3315 and returned at 152,977, having parked shallow enough that the $e^{\beta z}$ channel did not vanish. Both dynamical resurrections showed the ordered triple $t_{\text{death}} < t_{\text{area}} < t_{\text{res}}$, so the area law held 2/2. The gap between gate and return is the finite-β rise time, $\sim |z_{\text{park}}|/\text{drift}$, rather than the asymptotic $O(1/\beta)$ correction. The bottleneck was quantified as an event rate of ~ 0.1 per run, and that number set the entire sizing of the confirmatory.

Table 1: Confirmatory 4a.3: 400 runs, 100 per arm, $T = 0$, $H = 5 \cdot 10^5$. λ is the number of dynamical deaths per run. $\hat{p}_{\text{death}}$ conditions on all dynamical deaths in the arm; $\hat{p}_{\text{elig}}$ conditions on the deaths that occur inside the eligibility window $[0, 307{,}536]$, which is the only conditioning for which the return is observable within the horizon (for sp-β3 the eligible denominator is $N_{\text{elig}} = 30$ with 15 returns). Brackets are 95% Clopper–Pearson intervals, except the λ column, which carries a 95% bootstrap interval clustered by seed. ReLU is the validity control: any ground-truth return there would invalidate the run.

arm	dyn. deaths	λ/run [boot]	returns	$\hat{p}_{\text{death}}$	$\hat{p}_{\text{elig}}$
sp-β3	35	0.350 [0.220,0.500]	16	0.457 [0.288,0.634]	0.500 [0.313,0.687]
sp-β10	27	0.270 [0.160,0.390]	10	0.370 [0.194,0.576]	0.385 [0.202,0.594]
sp-β300	19	0.190 [0.100,0.290]	4	0.211 [0.061,0.456]	0.222 [0.064,0.476]
ReLU	17	0.170 [0.090,0.260]	0	0.000 [0.000,0.195]	0.000 [0.000,0.195]

Table 2: The pre-registered questions of 4a.3, with the condition that kills each one. All thresholds were derived programmatically from the rate pilot and frozen in `config_4a3.json` at commit `a8f8f8c`, before the run.

label	estimand	condition that kills	verdict
C1	$p(\text{ret} \mid \text{eligible})$, sp-$\beta$3	CP upper < 0.076, $N_{\text{elig}} \geq 10$	reproduces
C2	returns with area first, sp-β3	CP lower < 0.5, ≥ 6 returns	sustains
C3	rise gap against park depth	none (estimation)	not aggregated
C4	$p(\text{ret} \mid \text{death})$ against β	none (estimation)	monotone in β
V	returns in the ReLU arm	any single return	clean, 0/17

6.4 The area law reproduces under a pre-registered confirmatory

The confirmatory, `etapa-4a.3`, was pre-registered at commit `a8f8f8c` after two further adversarial rounds, with every design number derived programmatically from a 240-run rate pilot (commit `995a87c`); the run is commit `5a961cb`. Design grid: {sp-β3 (primary), sp-β10, sp-β300, ReLU} $\times$ $m \in \{4, 6\} \times 50$ held-out seeds (1900–1949, not used in any pilot), for 400 runs at $T = 0$ and $H = 5 \cdot 10^5$. The eligibility window is $[0, 307{,}536]$, that is $H - \tau$ with τ the $q_{0.9}$ return time of the pilot. The instrument is a single ground-truth clock with bit-exact parity to the pilot motor, and the area of the leader-datum residual is accumulated on that clock, so temporal ordering is structural and the only empirical failure mode of the area test is a return without area.

The verdicts (Tables 1 and 2 and Figure 4) are as follows.

- **C1, existence with rate, primary arm sp-β3: reproduces.** With $N_{\text{elig}} = 30$ against the floor $n^* = 10$, there are 15 eligible returns and $\hat{p}_{\text{elig}} = 0.500\,[0.313, 0.687]$, well above the kill threshold 0.076. Over all 35 dynamical deaths the same 16 returns give $\hat{p}_{\text{death}} = 0.457\,[0.288, 0.634]$; the two numbers are two conditionings of one count, not two estimates of one quantity.

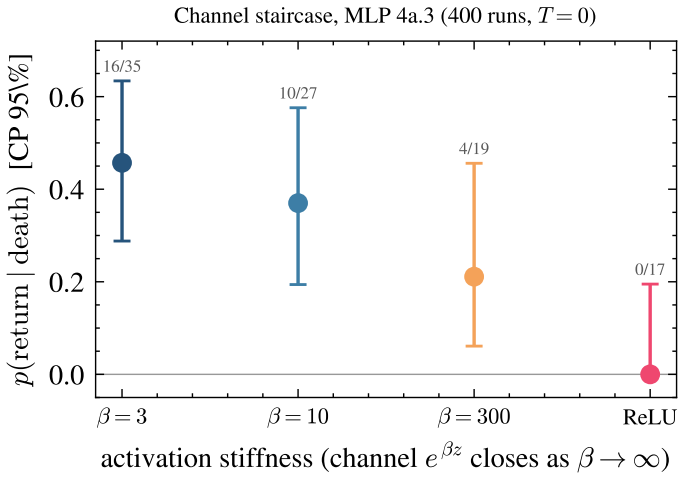


Figure 4: The channel staircase (C4, confirmatory 4a.3, 400 runs, $T = 0$). $\hat{p}_{\text{death}}$ is monotone in the activation stiffness and reaches a structural zero at hard ReLU, where the gradient vanishes identically; the smooth channel $e^{\beta z}$ closes as $\beta \to \infty$. The fractions above each point are returns over deaths and the error bars are 95% Clopper–Pearson intervals. Script `figuras_paper.py`.

- **C2, the area law, sp-β3: sustains.** The leader-drive area opens before the return in 13/16 dynamical resurrections, with lower Clopper–Pearson bound $0.544 > 0.5$ on ≥ 6 returns. The 3 returns without area are real and are counted against the law: they are returns driven off the leader datum, and they are an object of study.

- **C3, the rise gap: estimation, not aggregated.** C3 was pre-registered as an estimation-only quantity with no kill condition, relating the gap $t_{\text{res}} - t_{\text{area}}$ to the park depth z_{park} and to the channel $\sigma(\beta z_{\text{park}})$. Those quantities are recorded per episode by the instrument, but the frozen analysis script contains no aggregate for C3, and we report that rather than compute one after the fact.

- **C4, channel staircase: estimation, monotone in β.** $\hat{p}_{\text{death}}$ falls $0.457 > 0.370 > 0.211 > 0.000$ from $\beta = 3$ to hard ReLU, with the intervals of Table 1. The activation smoothness controls the return channel, and $\beta = 3$, the stiffness scale of the GELU family, is the arm with the highest return probability. This is the direct bridge to the plasticity practice of Section 4.

- **Validity: clean.** ReLU gave 0/17 dynamical returns, which is Proposition 11 measured with $N = 100$ runs.

The limits of the confirmatory are $T = 0$, one small architecture, a one-dimensional rich target, and $m \in \{4, 6\}$. The stochastic regime is a paper of its own (Section 6.1). The kill condition of C1 is intentionally near-unreachable at this N, a consequence of the frozen rule

that halves the pilot lower bound, so in practice C2 is the deciding test; this was signalled in the pre-registration rather than discovered afterwards.

6.5 Deviations from the pre-registered protocol

(i) None for 4a.3: the analysis frozen at `a8f8f8c` was run unchanged at `5a961cb`, and the validity control passed. (ii) The eligibility window $[0, 307{,}536]$, the kill threshold 0.076, the floor $n^* = 10$ for C1 and the floor of ≥ 6 returns for C2 were all derived programmatically from the rate pilot (`995a87c`) and frozen in `config_4a3.json` before the run; none was chosen after seeing the confirmatory data. (iii) C3 was pre-registered as estimation-only; its per-episode quantities are recorded but no aggregate is reported (Section 6.4). (iv) For the transformer, the pre-registered package was revised once before freezing: an earlier candidate was rejected in adversarial review for a confound between injection depth and the GELU barrier, and was replaced by the package of commit `1b45498`, which adds a depth-matched control. No run was executed under the rejected package.

6.6 The mechanism does not transfer unchanged to a GELU feed-forward block

The same question in a one-layer transformer, for the feed-forward block over an attended token, does not reduce to the MLP. The scoping run (exploratory, commit `d4e4c35`) exposed why, with two facts that reshaped the confirmatory:

- **S1, a GELU sign-flip barrier.** $\mathrm{GELU}'(z)$ changes sign at $z^* = -0.7518$, verified analytically: for $z < z^*$ it is negative, so a deep-dead unit is pushed away from the return by its own gradient. Self-resurrection is possible only in the shallow band $(-0.752, 0)$. The MLP's blind area port, which omits g', gets the sign wrong on 12/12 GELU episodes; the correct instrument uses the slope-weighted area $\mathcal{A}_{\mathrm{drive}}$. GELU is therefore genuinely outside the monotone-sigmoid class of Theorem 2.
- **S2, an upstream channel.** The input of the block, $h = e + \mathrm{attn}(e)$, is non-stationary, so a hard-ReLU unit whose own gradient is identically zero can still revive because h moves. With attention frozen the control is exact: the margin change is 0.000 and the own-move is 0. With attention training the own-move is still 0, yet the margin climbs by +0.69 on average and by up to +1.32. The MLP statement "dead ReLU is terminal" holds only when the input is fixed; the transformer adds a channel the MLP does not have.

A rate pilot of 160 runs (commit `2dba27d`) then fixed the design. Natural FFN death is ≈ 0, since units with $d = 16$ input do not die on their own, so the confirmatory is necessarily injection-conditional; the non-stationary channel chatters, forcing a hysteresis instrument with $\delta_h \approx 0.0476$ calibrated from data; and the return time is slow, $\tau \approx 60$–66k, forcing $H = 2 \cdot 10^5$. The upstream channel was already clean in the pilot: the difference between training and frozen attention was $+6/+6/+6/+8$ returns, and ReLU went from 0 to 8 with own-move 0.

The pre-registered protocol. A confirmatory run for the transformer was pre-registered at commit `1b45498`: 800 runs over {sp-β3, sp-β10, GELU, ReLU} $\times$ {attn-train, attn-frozen}$\times$ injection $\{-0.3, -2.0\} \times 50$ seeds 2000–2049, with $H = 2 \cdot 10^5$ and $T = 0$. Its results are not part of this note; the confirmatory run pre-registered at commit `1b45498` is reported in v2 of this note. We state the five pre-registered estimands here so that the protocol is on record ahead of the data.

- **E-A, validity.** Any sustained return in the relu-frozen arm will invalidate the run, since an own-gradient identically zero together with a fixed input makes death terminal. The pilot gave 0/40.
- **E-B, upstream channel.** We will estimate $\Delta = p_{\text{sust}}(\text{train}) - p_{\text{sust}}(\text{frozen})$ per arm, and will call the effect reproduced if the bootstrap lower bound is positive. Under the hysteresis instrument the test is stricter than in the pilot.
- **E-C, transfer of the theorem class.** For sp-β3 frozen under shallow injection we will estimate $p(\text{sustained return} \mid \text{injection})$, and will call the class transferred if there are at least $n^* = 6$ sustained returns. This is the direct analogue of C2 in the MLP.
- **E-D, GELU barrier.** For gelu-frozen we will compare deep injection (-2.0) against shallow (-0.3); the barrier predicts shallow far above deep in the own channel, with a depth-matched control against sp-β3. The pilot at -1.0 was a tie and did not decide.
- **E-E, β staircase.** We will estimate $p(\text{sustained return})$ across $\beta \in \{3, 10\}$, frozen and training, as an estimation with no kill condition.

The scoping verdict is itself a result: the MLP mechanism does not transfer directly to the GELU feed-forward block, because the sign-flip barrier obstructs self-resurrection and the upstream channel is a new route. The confirmatory tests the theorem class, softplus-β, on the block, and treats the GELU barrier and the upstream channel as estimands of their own.

7 Limitations

(i) The theorem is proved for one two-neuron instance; every constant is verified there and breadth is future work. **(ii)** The regime is $T = 0$ deterministic gradient flow throughout the theorem and the confirmatory; the stochastic gate is open and, where probed, falsifies the deterministic schedule (Section 6.1), since late captures recur past the deterministic window at a rate the reduced stochastic model under-predicts, leaving the reactivating unit's back-reaction as the leading candidate driver. **(iii)** Scale: the confirmatories are small networks ($m \leq 6$ MLP units, $m_{\text{ffn}} = 4$ block units) with rich one-dimensional targets, chosen so that a single unit can be tracked with a ground-truth clock. **(iv)** GELU is outside the monotone class, and transfer of the law to a transformer feed-forward block is stated as a protocol and left open. **(v)** The cone apex, of corank ≥ 2, where weight decay produces

genuine sliding toward the most singular stratum, is untouched. **(vi)** The independence of the law from the smoothing is proved for softplus and verified numerically for erf; it is not proved for the whole monotone-sigmoid class.

8 Conclusion

We derived an entry–exit law for the death and resurrection of a smoothed unit and verified each of its constants on a two-neuron instance. We measured the law in a small MLP under a pre-registered protocol: resurrection occurs, the residual area opens before the return in 13 of 16 cases, hard ReLU gives a structural zero, and the return probability is monotone in the activation stiffness. We provided evidence that the mechanism does not transfer unchanged to a GELU feed-forward block, where the derivative changes sign and where a non-stationary input opens a second route; that transfer is stated as a protocol and left open.

9 Reproducibility

All code, per-run manifests, raw episode files and aggregate JSON needed to reproduce every number and figure in this note are in a single Git repository. That repository is private at the time of writing, so the commit hashes below, while they resolve in it and fix the state of every experiment, are not independently checkable today; public release under the MIT licence is a precondition for submission, and we state this rather than let the reader discover it by clicking. All runs are deterministic on CPU with per-run manifests (SHA256, git hash, config), and pre-registration commits precede every confirmatory run in the repository history. No external data were used; every dataset in this note is generated by the scripts listed below.

- **Theorem constants and figures.** `06-notas/t2-prova/`: lemma verifications `passo0--6.py` and `prova_L1--L6.py`; figures via `figuras_draft.py`.
- **MLP mechanism (pilot).** `etapa4-mlp-populacao/pilotos-4a2/pressao/` (the *pressure* regime: sub-parametrized, plateau ~ 0.15); commit `b48b527`.
- **MLP confirmatory (4a.3).** `.../pressao/taxa/` (the event-rate pilot and the confirmatory built on it): pre-registration `a8f8f8c`, run `5a961cb`.
  `python run_4a3.py --workers 4` then `python analise_4a3.py`, which reads `results-4a3/` and `config_4a3.json`. Rate pilot `995a87c`. Staircase figure: `06-notas/paper/figuras_paper.py`.
- **Transformer (etapa-5).** `05-experimentos/etapa5-transformer/`: scoping `d4e4c35`, rate pilot `2dba27d`, confirmatory pre-registration `1b45498`.
  `python run_e5.py --workers 4` then `python analise_e5.py`.
- **Negatives.** `05-experimentos/NEGATIVOS.md` (the register of negative results) and the `etapa-4a` note record the invalidated instrument (commit `6411ac8`) and the

rejected redesign (`a413b19`); the falsified schedule is `p5-relu-morte-v2/` (commit `190d94a`).

Compute. Every run is single-threaded CPU on an Intel Core i9-9900K, with 4 concurrent workers through Python `multiprocessing` at `nice 19` and no GPU; the only dependency is `numpy` (Python 3.14, numpy 2.4). The 4a.3 confirmatory, 400 runs at $H = 5 \cdot 10^5$, took 4.9 h wall-clock, that is 19.5 CPU-hours at 175 s per run. The transformer protocol, 800 runs at $H = 2 \cdot 10^5$, is budgeted at ≈ 12.2 h. The preliminary and discarded runs, namely `etapa-4a` (600 runs), the `4a.2` redesign, the pressure pilot (64 runs) and the two rate pilots (240 and 160 runs), are of the same order again and are not otherwise reported.

Acknowledgments

This work received no funding and was carried out independently. The author declares no competing interests.

Data availability

All code, per-run manifests, raw episode files and aggregate JSON needed to reproduce every number and every figure in this note are held in the Git repository described in Section 9, at the commits cited there. That repository is not public at the time of writing; no external data were used, and all data are generated by the scripts listed in Section 9.

Declaration of LLM usage

Large language models were used throughout this work: to draft and copy-edit the text, to run adversarial reviews of each experimental design before its pre-registration (those referee reports are committed in the repository), to write the experiment and verification code, and to work out the derivations of Section A. Every lemma was then re-checked by an independent executable symbolic or numerical script, committed alongside it. The hypotheses, the pre-registered kill conditions, the decision to declare a design or a hypothesis dead, and the reading of the cited sources are the author's.

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A Proofs of the six lemmas

Each proof closes with the outcome of its committed executable verification. The scripts are symbolic where an identity is claimed and numerical where a rate or a constant is claimed; they are listed in Section 9. Throughout, $\sigma(s) = 1/(1+e^{-s})$, $w = e^{\beta z}$, $u = 1/w$ and $\Psi(u) = \log u + u$, and we use freely the elementary identities $\sigma(\beta z) = 1/(1+u)$, $\beta\, g_\beta(z) = \log(1+w)$ and $(1-\sigma(s))/\sigma(s) = e^{-s}$, all of which are verified symbolically in `passo0_verificacao.py`.

A.1 Normal form and additive quadrature

Proof of Lemma 5. Step 1 (margin equation, exact under (H3)). $\dot z = (x_j, 1)\cdot\dot\theta_A$, and the field on θ_A is $-\frac{1}{n}\sum_i \mathrm{res}_i\,\sigma(\beta z_i)(x_i,1)$. The term $i=j$ contributes $-c\,(g_\beta(z)-R)\,\sigma(\beta z)$, because $\mathrm{res}_j = g_\beta(z) - R$ by the definition (2) of R; the terms $i\neq j$ are bounded by $C_3' e^{-\beta\delta_1}$ by the scalar reduction of Lemma 10. Hence $\dot z = -c(g_\beta(z)-R)\sigma(\beta z) + O(e^{-\beta\delta_1})$, which in the gauge w and fast time $\tau = \beta t$ is (6).

Step 2 (exact chain rule). $\dot u = -\beta u\,\dot z$ and $\Psi'(u) = (1+u)/u$; together with $\sigma(\beta z) = 1/(1+u)$,

$$\frac{d\Psi}{dt} = \frac{1+u}{u}\cdot(-\beta u)\cdot\Big[-c\big(g_\beta(z)-R\big)\tfrac{1}{1+u}\Big] = c\,\beta\,\big(g_\beta(z)-R\big).$$

Step 3 (additive identity). Since $\beta\, g_\beta(z) = \log(1+e^{\beta z}) = \log(1+w)$, substitution gives $d\Psi/dt = -c\beta R + c\log(1+w)$, plus the (H3) error of Step 1 carried through the same factors, which is bounded by $C_3\beta e^{-\beta\delta_1}$. This is (7). No division by R occurs, so the interior zero of R forced by (H1)–(H2) through the intermediate value theorem is harmless.

Step 4 (bound on the correction, in three windows). On the episode $w = 1/u$ with $u = c\beta|\mathcal{A}(t)|(1+o(1))$ away from neighbourhoods of the endpoints (Lemma 8(i)). In the bulk, where $|\mathcal{A}| \geq a_0(\delta)$, one has $\log(1+w) \leq w \leq 1/(c\beta a_0)$, so that stretch contributes at most $H/(\beta a_0) = O(\beta^{-1})$. Near t_{hit} and t_{exit} the function $\mathcal{A}$ has transversal zeros, with derivatives $cR(t_{\mathrm{hit}}) \neq 0$ and $cR(t_{\mathrm{exit}}) \neq 0$, so $|\mathcal{A}(t)| \geq \kappa|t - t_*|$ locally and

$$\int \log(1+w) \;\lesssim\; \int_{1/\beta}^{\epsilon} \frac{ds}{c\beta\kappa s} = \frac{\log(\beta\epsilon)}{c\beta\kappa};$$

on the $O(\beta^{-1})$-wide stretches glued to the endpoints one has $w = O(1)$ and $\log(1+w) \le \log 2 + O(1)$, contributing $O(\beta^{-1})$. Summing the three windows gives $O(\log\beta/\beta)$ with constant $C_4 = 2/\big(c\min(|R(t_{\text{hit}})|, R(t_{\text{exit}}))\big) + H/a_0$. □

Remark 12 (consequence used downstream). Integrating (7) gives $\Psi(u(t)) = \Psi(u_0) - c\beta\mathcal{A}(t) + O(\log\beta/\beta)$ uniformly on the episode. This is the form that sustains the depth $u = c\beta|\mathcal{A}|(1+o(1))$ on compacts and the release at $t_{\text{exit}} + O(\beta^{-1})$, and it is what Lemmas 8 and 9 consume.

Verification. `prova_L1_verificacao.py` confirms the additive identity symbolically, and the integral correction with the normalized metric is stable in $[8.2, 10.6]$, below the theoretical $C_4 \approx 30$, for $\beta \in [3\cdot 10^2, 10^4]$.

A.2 The B-subsystem and the feedback

Proof of Lemma 6. Step 1 ((H4) makes the field linear). With the activation pattern constant, the B-field is $-G_B(\theta_B - \theta_B^*)$ with G_B the Gram matrix of the active data, hence C^∞ with Lipschitz constant $L_B = \lambda_{\max}(G_B)$, equal to 2.13 in the reference instance. Moreover $R^{\text{hyb}} = y_j - (x_j, 1)\cdot\theta_B$ is affine in θ_B and C^1 in t, with $\|\dot R^{\text{hyb}}\|_\infty \le \|(x_j,1)\|\,\|G_B\|\,\text{diam} =: C_R$ explicit.

Step 2 (coupling bound). The perturbation caused by A in res_i is $g_\beta(z_{A,i})$. Under (H3) the terms $i \ne j$ contribute $C_3 e^{-\beta\delta_1}$ and the term $i = j$ contributes $g_\beta(z)$; the Gram factors give a constant C_f, so $\|G_A(t)\| \le C_f\, g_\beta(z(t)) + C_3 e^{-\beta\delta_1}$.

Step 3 (integral Grönwall). Writing $\Delta = \theta_B^\beta - \theta_B^{\text{hyb}}$ we have $\dot\Delta \le L_B\Delta + \|G_A\|$, hence $\Delta(t) \le \int e^{L_B(t-s)}\|G_A(s)\|\,ds$, and therefore

$$\sup_{[t_{\text{cross}}, t_{\text{release}}]} \|\Delta\| \;\le\; e^{L_B H}\Big[\tfrac{C_f}{c\beta}\int c\log(1+w)\,dt + H\,C_3 e^{-\beta\delta_1}\Big].$$

Since $g_\beta = \beta^{-1}\log(1+w)$ and Lemma 5 gives $\int c\log(1+w)\,dt = O(\log\beta/\beta)$, the total is $O(\log\beta/\beta^2)$, which is (8). Consequently $R^\beta = R^{\text{hyb}} + O(\log\beta/\beta^2)$, which is the perturbation size consumed by Lemma 9. □

Verification. `prova_L2_verificacao.py` confirms that the field and R are affine with $L_B = 2.133$, that the margins of (H4) hold with $\min z_B = 0.39$, and that the refined rate holds: the metric $\sup\cdot\beta^2/\log\beta$ is stable in $[4.2, 5.7]$ over $\beta \in [3\cdot 10^2, 3\cdot 10^3]$. The exponent is what the metric discriminates: under a pure $O(\beta^{-1})$ the same metric would grow like $\beta/\log\beta$, by a factor of about 6.5 per decade, and it grew by a factor of 1.24.

A.3 The entry corner

Proof of Lemma 7. Step 1 (pre-corner with exponential weight). For $z \ge \delta_z$ one has $|g_\beta(z) - z| \le \beta^{-1}e^{-\beta z}$ and $1 - \sigma(\beta z) \le e^{-\beta z}$, so the inhomogeneity of the Grönwall comparison

against the ReLU flow is $h(t) \le Ce^{-\beta z^0(t)}$. Near the hit, transversality gives $z^0(t) = v\,(t_{\text{hit}} - t)(1 + O(\delta_z))$, hence

$$\int h\,dt \;\le\; C\int e^{-\beta v(t_{\text{hit}}-t)}\,dt \;\le\; \frac{C}{\beta v}.$$

Along the real trajectory the weight carries in addition the factor σ from the layer, and the exact value is $\int e^{-\beta z}dt = \frac{1}{\beta v}\int_0^\infty e^{-s}(1+e^{-s})\,ds\,(1+o(1)) = \frac{3/2}{\beta v}(1+o(1))$.

The weight is self-consistent by a monotone route, with no exponential fixed point. Since $e^{-\beta z}$ is decreasing in z, the flow whose inhomogeneity is evaluated at any sub-solution $z_- \le z^\beta$ majorizes the true inhomogeneity. Iterate: $z^{(0)} = z^0$, and let $z^{(k+1)}$ solve the problem with weight $e^{-\beta z^{(k)}}$. The iteration is monotone decreasing and bounded below by the ReLU sub-solution with slack $2C/(\beta v)$: the increment of each step is at most $C/(\beta v)\cdot \sup e^{\beta(z^{(k)}-z^{(k+1)})}$, and while $|z^{(k)} - z^{(k+1)}| \le 1/\beta$ that factor is at most e, a bound maintained by the size of the increment itself, by induction on k. The iteration therefore converges to z^β with $\Delta\theta(t_\delta) = O(1/(\beta v))$, without requiring a fixed-point inequality of the form $C_2e^{c'K} \le K$.

Step 2 (window-crossing identity). On $[0, \delta_z]$ the crossing speeds are $v_0(z) = v(1+O(\delta_z))$ for ReLU and $v_\beta(z) = v_0(z)\sigma(\beta z) + O(\beta^{-1})$ for softplus, so the difference of crossing times is

$$\begin{aligned}\int_0^{\delta_z}\Big(\frac{1}{v_\beta} - \frac{1}{v_0}\Big)dz &= \frac{1}{v}\int_0^{\delta_z}\frac{1-\sigma(\beta z)}{\sigma(\beta z)}\,dz\,(1+O(\delta_z)) \\ &= \frac{1}{v}\int_0^{\delta_z} e^{-\beta z}\,dz\,(1+O(\delta_z)) = \frac{1-e^{-\beta\delta_z}}{\beta v}(1+O(\delta_z)),\end{aligned}$$

where the middle equality is the exact identity $(1-\sigma(s))/\sigma(s) = e^{-s}$. With $w_0 = 1$ there is no further shift, since $\log w_0 = 0$. Adding Steps 1 and 2 gives $t_{\text{cross}} - t_{\text{hit}} = \frac{1}{\beta v}(1+O(\delta_z)+o(1))$, and the speeds are bounded on the window, so the state at crossing is within $O(\beta^{-1})$ of $\theta^\infty(t_{\text{hit}})$. □

Verification. `prova_L3_verificacao.py` confirms the two identities symbolically and both constants numerically: $K_{\text{teo}} = 1/v = 7.042$ against $K_{\text{emp}} = 7.127$ refitted from the measured crossing times, a deviation of 1.2%; and the exponentially weighted approach integral with the exact factor $3/2$ gives 10.563 in theory against 10.52 measured, below 1%.

A.4 Sticking and release for prescribed forcing

Proof of Lemma 8. Step 1 (monotone comparison). The right-hand side of $\dot\Psi = -c\beta R + c\log(1+w(\Psi))$ is decreasing in Ψ, because $w = 1/u$ and u is increasing in Ψ. The Chaplygin comparison principle therefore applies: $R_1 \le R_2$ implies $\Psi_1 \ge \Psi_2$, and piecewise-constant envelopes of R give sub- and super-solutions that are exactly integrable through the same primitive Ψ, hence with explicit constants.

Step 2 (depth, item (i)). Integrate the exact identity (7); the correction is bounded by Lemma 5, in the same three windows. On compact subintervals of $(t_{\text{hit}}, t_{\text{exit}})$ one has

u large, so $\Psi(u) = u(1 + o(1))$ and the depth law $u = c\beta|\mathcal{A}(t)|(1 + o(1))$ follows, that is $z \sim -\beta^{-1} \log \beta$.

Step 3 (release, item (ii)). At release Ψ returns to its entry value $\Psi(u_0)$, so the constant $\Psi(u_0)$ cancels and no spurious term survives; item (i) then gives $c\beta\mathcal{A}(t_{\text{release}}) = \text{corr}(t_{\text{release}})$. By (H2) the zero of $\mathcal{A}$ at t_{exit} is transversal, with $\mathcal{A}(t) = R(t_{\text{exit}})(t - t_{\text{exit}})(1 + o(1))$, and solving for t_{release} gives (10). The delay of the release is exactly the correction integral of Lemma 5, reparametrized by the exit speed.

Step 4 (perturbation, item (iii)). Apply the implicit function theorem to $\mathcal{A}_{\tilde{R}}(t) = 0$ with denominator $R(t_{\text{exit}}) \neq 0$, splitting the episode into the same three windows: near the hit $|\mathcal{A}| \geq |R(t_{\text{hit}})|s/2$ for $s = t - t_{\text{hit}}$ small, in the bulk $|\mathcal{A}| \geq a_0(\delta)$, and near the exit the zero is transversal. This gives the shift $\leq \delta\,(t_{\text{exit}} - t_{\text{hit}})/R(t_{\text{exit}}) + O(\delta^2)$. □

Remark 13 (the corollary that explains the coupled constant). In the coupled system the initial datum comes from the entry corner, so by Lemma 7 the episode starts with a delay K/β that cuts an initial negative area $\approx R(t_{\text{hit}})K/\beta$. Item (ii) then gives $\beta\,(t_{\text{release}} - t_{\text{exit}}^{\text{hyb}}) \to -K\,|R(t_{\text{hit}})|/R(t_{\text{exit}})$. With the theoretical $K = 1/(c|R(t_{\text{hit}})|)$ this is exactly $-1/(cR(t_{\text{exit}})) = -14.81$, the universal coefficient of (5); with the empirical $K = 7.13$ it is $-(7.13 \times 0.2272)/0.1080 = -15.0$, against -14.38 measured, a deviation of 4% whose residue is the positive correction of item (ii).

Verification. `prova_L4_verificacao.py` measures t_{release} and corr on the same run and tests $\Delta t = \text{corr}/(c\beta R_{\text{exit}})$: the formula closes to 0.2%/1.0%/9.5% at $\beta = 3 \cdot 10^2/10^3/3 \cdot 10^3$ on a grid of $2 \cdot 10^{-6}$, the last figure having collapsed from 35% to 9.5% under a fivefold grid refinement, which identifies the residue as discretization. The corollary of Remark 13 closes at -14.99 against -14.38 measured (4.2%).

A.5 The coupled bootstrap

Proof of Lemma 9. The argument is a continuation with strict margins.

Step 1 (the bootstrap property). For s in $[t_{\text{cross}}, t_{\text{exit}} - \varepsilon_\beta]$ let $\mathcal{P}_s$ be the property that, on $[t_{\text{cross}}, s]$, (a) $\sup \|\theta_B^\beta - \theta_B^{\text{hyb}}\| < 2M_2 \log\beta/\beta^2$ and (b) $z(t) < \beta^{-1} \log w_0 - \beta^{-1}$ for $t \geq t_{\text{cross}} + C_{\text{collar}}/\beta$. The collar is necessary: at t_{cross} the margin crosses $\beta^{-1} \log w_0$ continuously, so the layer can only be strict after an $O(\beta^{-1})$ entry window. The constant is $M_2 = e^{L_B H} C_f C_4/c$, explicit through the chain Lemma 6 composed with Lemma 5.

Step 2 (nonempty and open). Lemma 7 supplies the initial datum with slack: the state is within $O(\beta^{-1})$ in θ and the layer is reached transversally, so $\mathcal{P}_s$ holds for s just past the collar. All quantities in $\mathcal{P}_s$ are continuous in s, since the trajectories are C^1, so the strict inequalities propagate to a neighbourhood.

Step 3 (improvement, hence closed). Assume $\mathcal{P}_s$ with the constant $2M_2$. Since R is affine in θ_B (Lemma 6(a)), the forcing satisfies $R^\beta = R^{\text{hyb}} + \delta$ with $\|\delta\| \leq \text{Lip}_R \cdot 2M_2 \log\beta/\beta^2$. The sandwich of Lemma 8 with perturbed forcing then keeps $u(t) \geq c\beta(|\mathcal{A}(t)| - \|\delta\|H)/2 > u_0$ for $t \leq t_{\text{exit}} - \varepsilon_\beta$ with $\varepsilon_\beta = C_\varepsilon \log\beta/\beta$ and $C_\varepsilon = 4C_4/(cR_{\text{exit}})$, so the layer is maintained with a margin. With the layer in hand, the Grönwall of Lemma 6 is unconditional and returns M_2, half of what was assumed, for an explicit $\beta \geq \beta_0$. The threshold β_0 is finite and

traceable through the chain of constants, though we do not exhibit it. Crucially M_2 does not depend on the hypothesis $2M_2$, because the chain of constants is unconditional; this is what closes the bootstrap.

Step 4 (continuation). The set of s for which $\mathcal{P}_s$ holds is nonempty (Step 2), open (strict margins and continuity) and closed (the bounds pass to limits) in $[t_{\mathrm{cross}}, t_{\mathrm{exit}} - \varepsilon_\beta]$, hence it is the whole interval.

Step 5 (terminal window). On $[t_{\mathrm{exit}} - \varepsilon_\beta, t_{\mathrm{release}}]$ apply Lemma 8(ii)–(iii) with the forcing frozen at $R^{\mathrm{hyb}} + O(\log\beta/\beta^2)$. Three contributions add: the displacement caused by δ is $O(\log\beta/\beta^2)$ by item (iii); the correction delay is $+\mathrm{corr}/(c\beta R_{\mathrm{exit}})$ by item (ii); and the area cut at entry, $\int_{t_{\mathrm{hit}}}^{t_{\mathrm{hit}}+K/\beta} R \approx R(t_{\mathrm{hit}})K/\beta$, advances the exit by $-K|R(t_{\mathrm{hit}})|/(R(t_{\mathrm{exit}})\beta) = -1/(cR(t_{\mathrm{exit}})\beta)$, the arrival speed cancelling against $K = 1/(c|R(t_{\mathrm{hit}})|)$. Summing gives (11). □

Verification. `prova_L5_verificacao.py` checks that the two-term law reproduces the whole coupled sweep, with deviations 1.9%/0.7%/0.2%/0.1% for $\beta = 3\cdot 10^2 \ldots 10^4$, and that the universal coefficient is $1/(cR_{\mathrm{exit}}) = 14.815$; the chain of constants is exhibited with $M_2 = 440.8$ from the empirical $C_4 = 10.6$, against $M_2 \le 1233$ from the leading theoretical term $2/(cR^{\mathrm{hyb}}(t_{\mathrm{exit}})) = 29.6$ alone and $M_2 \lesssim 3300$ from the full theoretical $C_4 \approx 80$, which adds the bulk term H/a_0 with $a_0 \approx 0.02$ in the instance. The strict layer is confirmed at $\beta = 10^4$. One honest caveat: with the conservative C_ε the inner window of the layer check is empty for $\beta \le 3\cdot 10^3$, so the check is vacuous there, and at those β the maintenance of the layer is implied only by the detection of the release itself.

A.6 The exit corner, branch selection and the scalar reduction

Proof of Lemma 10. Step 1 (item (i)). Mirror Lemma 7, replacing v by $v' = cR^{\mathrm{hyb}}(t_{\mathrm{exit}}) > 0$, which is transversal by (H2). The crossing identity is the same, and the weight is $\frac{3/2}{\beta v'}$.

Step 2 (item (ii), forward uniqueness and branch selection). At the restart point the active field is locally Lipschitz on the side $z \ge 0$ and $\dot z(t_{\mathrm{exit}}^+) = v' > 0$, so the solution enters $\{z > \epsilon t\}$ and Picard uniqueness holds forward from any $t > t_{\mathrm{exit}}$. The only Filippov alternative [7], namely remaining in $z \le 0$ where the field is identically zero, is incompatible with the limit of the θ^β, since Lemma 9 places the release at $t_{\mathrm{exit}} + O(\beta^{-1})$. The selection of the activating branch is therefore proved by Lemma 9 and not postulated in Definition 1.

Step 3 (item (iii), gluing). The initial datum of the gluing satisfies $\|\theta^\beta(t_{\mathrm{release}}) - \theta^\infty(t_{\mathrm{exit}})\| = O(\beta^{-1}\log\beta)$: the position of θ_A in the layer contributes $O(\log\beta/\beta)$, θ_B contributes the rate of Lemma 6, and the time displacement $O(\beta^{-1})$ contributes through bounded speeds. In the active interior the two fields are exponentially close, so classical Grönwall on $[t_{\mathrm{release}}, H]$ keeps the gap at $O(\log\beta/\beta)$.

Step 4 (scalar reduction). The field on θ_A is $-\frac{1}{n}\sum_i \mathrm{res}_i\,\sigma(\beta z_i)(x_i, 1)$. Under (H3), that is $z_{A,i} \le -\delta_i < 0$ for $i \ne j$ on the limit episode, the terms $i \ne j$ are bounded by $Ce^{-\beta\delta_i}$ and are absorbed by the Grönwall estimates without changing any $O(\beta^{-1})$ rate. The term $i = j$ is rank one along $(x_j, 1)$, so the component of θ_A tangential to the face is frozen at the exponential rate, while the normal component is exactly the scalar problem of Lemmas 5

to 9. Hypothesis (H3) is an open condition on the data and the initialization, and it is verified numerically in the instance with an explicit margin δ_1. □

Verification. `prova_L6_verificacao.py` measures the gluing directly: for $\beta \in \{10^3, 3 \cdot 10^3\}$ the supremum of $\|\theta^\beta - \theta^{\mathrm{relu}}\|$ over $[t_{\mathrm{release}} + 0.05, H]$ is $1.6 \cdot 10^{-3}$ and $5.5 \cdot 10^{-4}$, so the metric $\sup \cdot \beta / \log \beta$ is stable at 0.21–0.23, confirming the rate $O(\log \beta / \beta)$. `passo6_verificacao.py` confirms the scalar reduction: (H3) holds with $\delta_1 = 0.508$ and the tangential drift is below 10^{-10}, machine zero, at both values of β, consistent with $e^{-\beta\delta_1} \sim 10^{-65}$.

The debugging of this script is worth recording, because it exhibits the alternative branch. The first version of the reference restarted the ReLU flow at $z_A = -1.1 \cdot 10^{-6}$, an overshoot of one Euler step of the pre-hit loop, and fell into the frozen Filippov branch, with a constant gap of ≈ 0.039. The correct reference requires exact projection onto the face together with an activating nudge larger than the overshoot. Branch selection is a real condition of the statement.